

Introduction to abelian categories

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Abstract

We develop the theory of abelian categories, motivated by homological algebra in Mod_R . We then develop homology and the science of diagram chasing. If time permits, we will close with a discussion of derived functors.

Contents

1	Introduction	1
2	$\mathcal{A}b$ -categories	1
3	Additive categories	3
4	Abelian categories	4
5	Examples and constructions in abelian categories	6

1 Introduction

Throughout the Colloquium (I hope I spelled that right) we have discussed the theory of homology and its applications at length, but we have always taken our chain complexes to have R -modules as their entries, or sometimes abelian groups or vector spaces, both of which are of course special cases of R -modules. A natural question is whether we can construct chain complexes in more general categories, broadening our class of objects to which we can associate a theory of homological algebra.

However, homological algebra has a large number of requirements, both obvious and non-obvious. For example, given a chain complex $C_\bullet$, the n th homology $H_n(C_\bullet)$ is the quotient $\text{Im } d_{n-1} / \text{Ker } d_n$. Even just to construct homology, we need to define kernels and images (we do not need quotients, as we will see). Kernels don't even exist in many mundane categories, such as Set and Top . Additionally, it may not be clear how to abstract images and kernels to non-concrete categories, since they are fundamentally constructed using the underlying sets of objects.

To this end, we develop the theory of *abelian categories* as a “good place to do homological algebra”. These categories were first introduced by Grothendieck in Tohoku for this very purpose, and nearly all constructions in homological algebra can be traced back to the theory of abelian categories in some way or another.

Throughout this talk, we will develop the theory of abelian categories using this motivation. We will then construct homology in these categories, and introduce some of the basic theory of diagram chasing. If time permits, we will close with a discussion of derived functors.

Convention 1.1. Given a category $\mathcal{C}$, we write $x, y \in \mathcal{C}$ to denote that x, y are objects in $\mathcal{C}$, and $\mathcal{C}(x, y)$ to denote the hom-set of morphisms $x \rightarrow y$ in $\mathcal{C}$. Important categories are denoted with capital letters in script and lowercase letters in roman (eg. Top , Set , sSet). Functors are denoted by capital roman letters $F : \mathcal{C} \rightarrow \mathcal{D}$, unless there are other conventions (eg. quotient functors denoted by π). We will always denote the group operation in abelian groups by addition, and will always denote the group operation in non-abelian groups by multiplication.

2 $\mathcal{A}b$ -categories

The prototypical example of an abelian category is the category $\mathcal{A}b$ of abelian groups, which can be thought of as $\mathbb{Z}$ -modules. This is where abelian categories get their namesake. We will however use

Mod_R as our foundational example throughout these notes since they are slightly more general, but for those who are unfamiliar with module theory, every single thing we discuss can be replaced by “ k -vector space”, where k is your favorite field (eg. $\mathbb{R}, \mathbb{C}, \mathbb{Q}$).

Let $\mathcal{C}$ be a category. Before we can even begin to define homology of chain complexes in $\mathcal{C}$, we first have to construct chain complexes in $\mathcal{C}$. We might naïvely think this is fairly simple. A chain complex $C_\bullet$ in Mod_R is simply a collection $\{C_n\}_{n \in \mathbb{Z}}$ of R -modules, and R -linear maps $d_n : C_n \rightarrow C_{n-1}$ such that $d_{n-1} \circ d_n = 0$. So to abstract this, we simply take the objects C_n to be objects of $\mathcal{C}$, and the maps d to be morphisms $d_{n-1} : C_n \rightarrow C_{n-1}$ in $\mathcal{C}$, which compose to 0.

However, we have already run into an issue. What does it mean for a morphism to be “0” in an arbitrary category? In Mod_R , the 0 map $0 : M \rightarrow N$ maps each $a \in M$ to the identity $0 \in N$. In arbitrary categories, we cannot argue elementwise, which means we need to find a different, yet equivalent, characterization of these maps which does not invoke elements.

We begin by studying the hom-set $\text{Mod}_R(M, N)$ of R -linear homomorphisms $M \rightarrow N$. A particularly useful fact is that these have the structure of an *abelian group*. Given R -linear maps $\varphi, \psi : M \rightarrow N$, we define $\varphi + \psi$ pointwise, meaning for any $m \in M$, we define $(\varphi + \psi)(m) = \varphi(m) + \psi(m)$. This map is indeed an R -linear map, since for all $m, n \in M$,

$$\begin{aligned} (\varphi + \psi)(m + n) &= \varphi(m + n) + \psi(m + n) = \varphi(m) + \varphi(n) + \psi(m) + \psi(n) \\ &= \varphi(m) + \psi(m) + \varphi(n) + \psi(n) = (\varphi + \psi)(m) + (\varphi + \psi)(n). \end{aligned}$$

The first step follows by definition of $\varphi + \psi$, the second by R -linearity of these maps, the third by commutativity of $+$ in R -modules, and the fourth again by definition of $\varphi + \psi$. R -linearity is also simple: for any $r \in R$,

$$(\varphi + \psi)(rm) = \varphi(rm) + \psi(rm) = r\varphi(m) + r\psi(m) = r(\varphi + \psi)(m).$$

Associativity is easily shown as well, and one simply defines the inverse $-\varphi$ of φ as $x \mapsto -1 \cdot \varphi(x)$, where -1 is the additive inverse of 1 in R . A more interesting fact is that the 0 map is the identity of the group:

$$(\varphi + 0)(m) = \varphi(m) + 0(m) = \varphi(m) + 0 = \varphi(m).$$

This gives us a new way to characterize the 0 morphism: it is the identity element of the abelian group Mod_R .

However, arbitrary categories don’t necessarily have abelian group structures on their hom-sets. This means we must specialize to what are called *Ab-categories* or *preadditive categories*.

Definition 2.1. An *Ab-category* is a category enriched over *Ab*. More explicitly, an *Ab-category* $\mathcal{C}$ is a category wherein every hom-set $\mathcal{C}(x, y)$ has an abelian group structure, such that addition in hom-sets distributes over composition.

Many books prefer to say that addition is *bilinear* with respect to composition, instead of simply saying it distributes. Although they are indeed equivalent notions, I see no reason to use more confusing terminology than needed. This condition is required to ensure that the category behaves well with respect to this abelian group structure.

Remark 2.2. For those who know some enriched category theory, you will note that when working with *Ab-categories*, we make use of the isomorphism $\text{Ab}(\mathbb{Z}, A) \cong A$ to identify the category with its underlying category. We thus say an *Ab-category* is simply a category where morphisms can be added together, which is equivalent in this way to the usual notion of an enriched category.

We may now realize the definition of a zero morphism in an arbitrary *Ab-category*:

Definition 2.3. Let $\mathcal{C}$ be an $\mathcal{A}b$ -category, and $x, y \in \mathcal{C}$. The zero morphism $0 : x \rightarrow y$ is the identity element of the abelian group $\mathcal{C}(x, y)$.

One reason we require addition to distribute over composition in $\mathcal{A}b$ -categories is the following fact. In Mod_R , composing by the zero morphism yields the zero morphism, meaning for any R -linear map φ ,

$$\varphi \circ 0 = 0, \quad 0 \circ \varphi = 0.$$

This fact generalizes to $\mathcal{A}b$ -categories, using distributivity of addition over composition. Let $\varphi : x \rightarrow y$ in an $\mathcal{A}b$ -category $\mathcal{C}$. Then

$$\begin{aligned} \varphi \circ 0 &= \varphi \circ (0 + 0) = (\varphi \circ 0) + (\varphi \circ 0) \\ \implies (\varphi \circ 0) - (\varphi \circ 0) &= (\varphi \circ 0) + (\varphi \circ 0) - (\varphi \circ 0) \implies 0 = \varphi \circ 0. \end{aligned}$$

We may now at the very least construct chain complexes in an $\mathcal{A}b$ -category. However, homological algebra requires *homology* (hence the name), and in fact requires even more to be able to study homology in any practical way, so we have a long way to go.

3 Additive categories

The next step up from an $\mathcal{A}b$ -category is an *additive category*. An extremely important object in homological algebra is the trivial R -module 0 . The trivial R -module is used as the “start” and “end” of many chain complexes, including short exact sequences, and also characterizes when there is no homology in a complex. To have a good theory of homological algebra, we should require that our categories have a zero object. Luckily, regular category theory already provides a good definition.

Definition 3.1. A *zero object* or *null object* of a category $\mathcal{C}$ is an object $0 \in \mathcal{C}$ which is both initial and terminal.

Indeed, $0 \in \text{Mod}_R$ is both initial and terminal, so this is a good definition. Moreover, the unique maps $0 \rightarrow M$ and $M \rightarrow 0$ must be the zero map since abelian groups must have an identity, which agrees with Mod_R further.

The next definition we introduce is largely motivated by the structure of Mod_R , and less so by homological algebra. It may not be immediately clear how this is relevant in the study of homology, but many extremely important constructions and theorems in homological algebra make use of the notion of a *biproduct*. To motivate this notion, we present a fascinating theorem.

Proposition 3.2. Let $\mathcal{C}$ be an $\mathcal{A}b$ -category, and $x, y \in \mathcal{C}$. If a product $x \times y$ of x, y exists, then a coproduct $x \amalg y$ also exists, and there is an isomorphism $x \times y \cong x \amalg y$. In particular, finite products and coproducts coincide.

The proof is nontrivial, and can be found in [nLa25] proposition 2.1.

Definition 3.3. Let $\mathcal{C}$ be a category. Two objects $x, y \in \mathcal{C}$ have a *biproduct* when their product and coproduct coincide. We then call it a *biproduct* and denote it by $x \oplus y$.

In practice, we rarely actually call the biproduct a biproduct, and simply call it a product. We also rarely use the notation $\oplus$ outside of *additive categories*, which we will shortly introduce.

The category Mod_R has finite products and coproducts, and by proposition 3.2 they coincide. The product of two R -modules $M \times N$ is the same as their coproduct $M \amalg N$, so we call it a *direct sum* and denote it by $M \oplus N$. Direct sums are extremely useful in the theory of R -modules, and it would be nice if our $\mathcal{A}b$ -categories had finite biproducts as well. This motivates definition 3.4.

Definition 3.4. An $\mathcal{A}b$ -category $\mathcal{C}$ is an *additive* category if it has a zero object and binary biproducts. Equivalently, it must have a zero object and binary products (or coproducts), since products *are* coproducts in $\mathcal{A}b$ -categories by proposition 3.2.

4 Abelian categories

We are now prepared to begin defining *abelian categories*. We already know of one requirement they must satisfy: an abelian category must be an additive category. This section will look for other properties they should satisfy in order for us to have a good theory of homological algebra.

The first requirement we must make is the existence of kernels. Kernels are usually constructed as *subsets* of R -modules or groups, so to generalize their definition to arbitrary categories, we must explore different characterizations of kernels. The most natural characterization is by way of a universal property.

To see how kernels arise by a universal property, let $\varphi : M \rightarrow N$ be R -linear. Then the kernel $\text{Ker } \varphi$ is a *submodule* of M . This means that there is an inclusion map $i : \text{Ker } \varphi \hookrightarrow M$ which is also R -linear. This inclusion map satisfies an interesting property. Every $m \in \text{Ker } \varphi$ gets mapped to 0 by φ by the definition of a kernel, meaning $\varphi \circ i = 0$, since the inclusion simply restricts φ to the kernel. Additionally, the kernel is *universal* with respect to this property, meaning for any $\psi : P \rightarrow M$ such that $\varphi \circ \psi = 0$, the map ψ factors uniquely through the kernel of φ . Alternatively, given any diagram of solid arrows

$$\begin{array}{ccccc}
 & & 0 & & \\
 & \searrow & \text{---} & \nearrow & \\
 P & \xrightarrow{\psi} & M & \xrightarrow{\varphi} & N \\
 & \searrow & \uparrow i & & \\
 & & \text{Ker } \varphi & &
 \end{array}$$

there is a unique dashed arrow as indicated, rendering the diagram commutative.

Since zero morphisms exist in $\mathcal{A}b$ -categories, we have an immediate generalization of the notion of kernel to $\mathcal{A}b$ -categories.

Definition 4.1. Let $\mathcal{C}$ be an $\mathcal{A}b$ -category, and $\varphi : x \rightarrow y$ a morphism in $\mathcal{C}$. A kernel $\text{Ker } \varphi$, if it exists, is an object of $\mathcal{C}$, together with a morphism $\text{ker } \varphi : \text{Ker } \varphi \rightarrow x$, such that for any diagram of solid arrows

$$\begin{array}{ccccc}
 & & 0 & & \\
 & \searrow & \text{---} & \nearrow & \\
 z & \xrightarrow{\quad} & x & \xrightarrow{\varphi} & y \\
 & \searrow & \uparrow \text{ker } \varphi & & \\
 & & \text{Ker } \varphi & &
 \end{array}$$

there is a unique dashed arrow as indicated, rendering the diagram commutative.

Note the interesting shift in notation. The morphism $\text{Ker } \varphi \rightarrow x$ is now denoted by $\text{ker } \varphi$. Since most of the structure in arbitrary categories is carried by the morphisms, when working outside of Mod_R , it usually makes sense to refer to the *inclusion morphism* as the kernel of φ , hence our notation. We use the capital K in $\text{Ker } \varphi$ to denote the object, and the lowercase k in $\text{ker } \varphi$ to denote the morphism.

Remark 4.2. Kernels also have a dual, called the *cokernel*. Given some $\varphi : x \rightarrow y$ in an $\mathcal{A}b$ -category $\mathcal{C}$, a *cokernel* $\text{Coker } \varphi$ of φ is an object, together with a morphism $\text{coker } \varphi : y \rightarrow \text{Coker } \varphi$ with $\text{coker } \varphi \circ \varphi = 0$, such that for any diagram of solid arrows

$$\begin{array}{ccccc}
 & & \text{Coker } \varphi & & \\
 & & \uparrow \text{coker } \varphi & \searrow & \\
 x & \xrightarrow{\varphi} & y & \xrightarrow{\quad} & z \\
 & \searrow & \downarrow 0 & \nearrow & \\
 & & & &
 \end{array}$$

there is a unique dashed arrow as indicated, rendering the diagram commutative.

We will also need a good notion of an *image* object $\text{Im } \varphi$ for any morphism $\varphi : x \rightarrow y$ in an additive category, and we will also need a way to construct the *quotient* of the image and the kernel, such that we can obtain homology. Let's begin by considering quotients.

Let $\varphi : M \rightarrow N$ be R -linear. The cokernel $\text{Coker } \varphi$ can be constructed as the quotient module

$$\text{Coker } \varphi \cong \frac{N}{\text{Im } \varphi}.$$

This follows since this object is initial with respect to mapping $\text{Im } \varphi$ to 0. Note that since this is constructed as a quotient module, we might want to think of cokernels as quotients, giving us an abstraction of cokernels. Since $\text{Im } d_n \subseteq \text{Ker } d_{n-1}$ in a chain complex $C_\bullet$, we can thus view the homology $H_n(C_\bullet)$ as the cokernel of d_n , viewed as a map $C_n \rightarrow \text{Ker } d_{n-1}$, rather than a map $C_n \rightarrow C_{n-1}$.

This gives a purely diagram-theoretic construction of homology. Let $C_\bullet$ be a chain complex, and since $d_{n-1}d_n = 0$, the map d_n factors through the kernel as indicated by the left triangle

$$\begin{array}{ccccc}
 C_n & \xrightarrow{d_n} & C_{n-1} & \xrightarrow{d_{n-1}} & C_{n-2} \\
 & \searrow d'_n & \uparrow \ker d_{n-1} & & \uparrow \\
 & & \text{Ker } d_{n-1} & \xrightarrow{\text{coker } d'_n} & \text{Coker } d'_n
 \end{array}$$

By commutativity of the left triangle, $d_{n-1} \circ \ker d_n \circ d'_n = 0$, and thus the second two maps factor through the cokernel. By our knowledge of kernels in Mod_R , we know that $\text{Im } d'_n = \text{Im } d_n$, when viewed as submodules of C_{n-1} . Thus, if we demand that kernels and cokernels exist, we are guaranteed a purely categorical construction of homology by taking $H_n(C_\bullet) := \text{Coker } d'_n$.

There is one final requirement of abelian categories, which is motivated by a particular property of Mod_R that characterizes monics and epis. A monic map $M \rightarrow N$ is simply an injective map, meaning it identifies M with a subspace $M \cong \text{Im } \varphi \subseteq N$. Since all subspaces are kernels, $\text{Im } \varphi = \text{Ker } \psi$ for some ψ , meaning M can be identified with the kernel of ψ , and thus φ is a *kernel*. Thus, monics are kernels. Conversely, epis coincide with cokernels. This property makes kernels and cokernels extremely well-behaved, and will help us develop a lot of important theory, such as a categorical definition of images and coimages. We thus demand that abelian categories have this property as well.

We have now identified all the properties an abelian category must have, in order to have a well-behaved theory of homology. We summarize with

Definition 4.3. Let $\mathcal{C}$ be an $\mathcal{A}b$ -category. Then $\mathcal{C}$ is an *abelian category* if

1. $\mathcal{C}$ is additive (definition 3.4);
2. every morphism in $\mathcal{C}$ has a kernel and a cokernel;
3. every monic arrow is a kernel, and every epi arrow is a cokernel.

5 Examples and constructions in abelian categories

We begin this section with some examples of abelian categories. Clearly, Mod_R is an abelian category, and thus so is $\mathcal{A}b$, since we modeled them after these categories. The following construction gives another example.

Construction 5.1. Let $\mathcal{A}$ be an abelian category. The category $\text{Ch}(\mathcal{A})$ of *chain complexes* in $\mathcal{A}$ is defined by taking objects to be chain complexes $x_\bullet \in \text{Ch}(\mathcal{A})$, and morphisms $f_\bullet : x_\bullet \rightarrow y_\bullet$ are collections of functions $\{f_n\}_{n \in \mathbb{Z}}$ which commute with differentials in the sense that the diagram

$$\begin{array}{ccccccc}
 \dots & \xrightarrow{d_{n+2}} & x_{n+1} & \xrightarrow{d_{n+1}} & x_n & \xrightarrow{d_n} & x_{n-1} \xrightarrow{d_{n-1}} \dots \\
 & & \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} \\
 \dots & \xrightarrow{d_{n+2}} & y_{n+1} & \xrightarrow{d_{n+1}} & y_n & \xrightarrow{d_n} & y_{n-1} \xrightarrow{d_{n-1}} \dots
 \end{array}$$

commutes. More precisely,

$$f_{n-1} \circ d_n^x = d_n^y \circ f_n.$$

Composition and identity are given pointwise.

Proposition 5.2. *The category $\text{Ch}(\mathcal{A})$ is abelian.*

We don't offer a full proof, but rather we note that $\text{Ch}(\mathcal{A})$ is simply a full subcategory of $\text{Fun}(\mathbb{Z}, \mathcal{A})$, and the relevant universals exist in the functor category and are defined pointwise by [ML98]. Some wrangling is needed to show the universals are indeed in $\text{Ch}(\mathcal{A})$, and that the abelian structure is promoted.

Another example of an abelian category in particular use in algebraic geometry is as follows. Given an abelian category $\mathcal{A}$ and a topological space X (more generally, a site), the category $\text{Shf}_{\mathcal{A}}(X)$ of sheaves on X with values in $\mathcal{A}$ is an abelian category. This means categories such as the category $\text{Shf}_{\mathcal{A}b}(X)$ of sheaves of abelian groups are useful, since one can immediately do homological algebra with sheaves.

References

- [ML98] Saunders Mac Lane. *Categories for the Working Mathematician*, volume 5 of *Graduate Texts in Mathematics*. Springer, second edition, 1998.
- [nLa25] nLab authors. additive category. <https://ncatlab.org/nlab/show/additive+category>, September 2025. Revision 46.